TEXORA · LECTURE SHEET

Entropy

The state function that gives the second law its arithmetic: definition through reversible heat, entropy changes in the standard processes and phase changes, the entropy of the universe and irreversibility, Boltzmann's statistical meaning, and the third law.

PHY-1203 · PHYSICS-II · TOPIC 5 · LEVEL 1 TERM 2

1. Definition — entropy as a state function

The Carnot cycle showed that for a reversible cycle $\oint dQ/T = 0$ exactly. A quantity whose cyclic integral vanishes is a **state function**, so Clausius defined **entropy** S by

$$\Delta S = \int (dQ_{rev}/T), \quad \text{unit J K}^{-1}$$

The subscript matters: to *compute* an entropy change between two states, integrate along any **reversible** path joining them — the answer then holds for every path, reversible or not, because S depends only on the endpoints. Entropy is extensive (two systems' entropies add) and, with energy, is one of the two great bookkeeping quantities of physics.

2. Entropy changes — the standard calculations

CHANGE	ΔS
Isothermal expansion of ideal gas (T const)	$nR \ln(V_2/V_1)$
Heating at constant c ($T_1 \rightarrow T_2$)	$mc \ln(T_2/T_1)$ (or $nC \ln(T_2/T_1)$)
Phase change at constant T (melting, boiling)	mL/T
Reversible adiabatic	0 — the process is *isentropic*
Free expansion of ideal gas	$nR \ln(V_2/V_1)$ — computed along a reversible isothermal substitute path

WORKED EXAMPLES

One mole doubling its volume isothermally at any T: $\Delta S = R \ln 2 = 8.314 \times 0.693 =$ **5.76 J K⁻¹**. Melting 1 kg of ice at 273 K ($L = 3.34 \times 10^5$ J kg⁻¹): $\Delta S = 3.34 \times 10^5/273 =$ **1,223 J K⁻¹** —

> melting is a huge entropy purchase, made at a single temperature.

3. Entropy of the universe and irreversibility

For any real process, system + surroundings obey the **entropy principle**:

$$\Delta S_{universe} \geq 0, \quad \text{equality only for a reversible process}$$

This one inequality *is* the second law in working clothes. Check it on heat conduction: 1,000 J flowing from a 400 K reservoir to a 300 K reservoir gives $\Delta S = -1{,}000/400 + 1{,}000/300 = -2.50 + 3.33 =$ **+0.83 J K⁻¹** — positive, so the flow is allowed, and the reverse flow (which would give −0.83) is forbidden. A warmer check with mixing: 1 kg of water at 400 K mixed with 1 kg at 300 K settles at 350 K; $\Delta S_{hot} = 4{,}186 \times \ln(350/400) = -559$ J K⁻¹, $\Delta S_{cold} = 4{,}186 \times \ln(350/300) = +645$ J K⁻¹, total **+86 J K⁻¹** — the mixture can never un-mix itself, and now you can say by exactly how much it cannot.

4. The statistical meaning — Boltzmann

Boltzmann identified entropy with the number W of microscopic arrangements (microstates) realising a macrostate:

$$S = k \ln W$$

(k = Boltzmann's constant; the formula is engraved on his tombstone.) Entropy then measures microscopic disorder or, better, *spread*: energy and matter distributed over more accessible arrangements. It also re-derives the gas results: doubling a gas's volume doubles each molecule's accessible positions, multiplying W by 2^N, so $\Delta S = k \ln 2^N = Nk \ln 2 = nR \ln 2$ — the same 5.76 J K⁻¹ per mole as the heat calculation. Heat is the clumsy accountant; counting is the truth underneath.

5. T–S diagrams and the third law

On a temperature–entropy diagram the area under a reversible path equals the heat absorbed ($Q = \int T\,dS$), and a cycle's enclosed area equals its net work — the Carnot cycle becomes a rectangle, a picture worth memorising. The **third law** (Nernst) closes the scale: the entropy of a perfect crystal approaches zero as $T \to 0$ K, giving entropies an absolute zero point and explaining why heat capacities vanish near absolute zero — and why absolute zero itself is unattainable in finitely many steps.

6. Key points

- $\Delta S = \int dQ_{rev}/T$; S is a state function — always compute along a reversible path, apply to any path.

- Standard results: isothermal gas nR ln(V_2/V_1); heating mc ln(T_2/T_1); phase change mL/T; reversible adiabatic zero.
- ΔS_universe ≥ 0 decides the direction of every process; irreversible processes manufacture entropy.
- S = k ln W ties entropy to microstate counting and reproduces the thermodynamic formulas.
- T–S area = heat (or net work for a cycle); third law: S → 0 at 0 K for a perfect crystal.

7. Practice

Answer in your own words; brief model answers follow.

1. Two moles of an ideal gas expand isothermally at 350 K to three times the volume. Find ΔS of the gas and of the universe (the expansion is reversible).
2. Half a kilogram of ice melts at 0 °C (L = 3.34 × 10^5 J kg^{-1}). Find the entropy change of the ice/water.
3. Two kilograms of water are heated from 20 °C to 80 °C at constant pressure (c = 4,186 J kg^{-1} K^{-1}). Find ΔS.
4. Heat of 500 J flows spontaneously from a body at 500 K to one at 250 K. Compute ΔS_universe and state what the sign tells you.
5. A free expansion doubles a mole of gas's volume. Compute ΔS two ways: thermodynamically, and statistically from S = k ln W.
6. Why must a real refrigerator's entropy balance be checked on the universe, not the fridge, to test the second law? Sketch the four entropy changes involved.

Answer key

1. ΔS_gas = nR ln 3 = 2 × 8.314 × 1.099 = 18.3 J K^{-1}. Reversible: the surroundings lose heat at the same temperature at the same rate, ΔS_surr = −18.3 J K^{-1}, so ΔS_universe = 0 — the signature of reversibility.
2. ΔS = mL/T = 0.5 × 3.34 × 10^5/273 = 167,000/273 ≈ 612 J K^{-1}.
3. ΔS = mc ln(T_2/T_1) = 2 × 4,186 × ln(353/293) = 8,372 × 0.186 ≈ 1,560 J K^{-1} ≈ 1.56 kJ K^{-1}. (Kelvin, always.)
4. ΔS_universe = −500/500 + 500/250 = −1 + 2 = +1 J K^{-1}. Positive: the flow is spontaneous and irreversible; the reverse flow would give −1 J K^{-1} and is forbidden.
5. Thermodynamic: ΔS = nR ln 2 = 5.76 J K^{-1} (via the reversible isothermal path between the same states). Statistical: each molecule gains 2× positions, W multiplies by 2^N, ΔS = Nk ln 2 = (1 mol × R) ln 2 = 5.76 J K^{-1}. Identical — the two definitions agree.
6. Because a fridge's contents can lose entropy (they get colder and more ordered) without violating anything — entropy is not conserved, only non-decreasing for the whole. The four changes:

food/interior $\Delta S < 0$; kitchen air receiving rejected heat $\Delta S > 0$; the work source (power station) $\Delta S > 0$ from fuel burnt; refrigerant itself returns to its state each cycle, $\Delta S = 0$. Sum ≥ 0, with equality only for a Carnot refrigerator.
